The Power of Vaccination

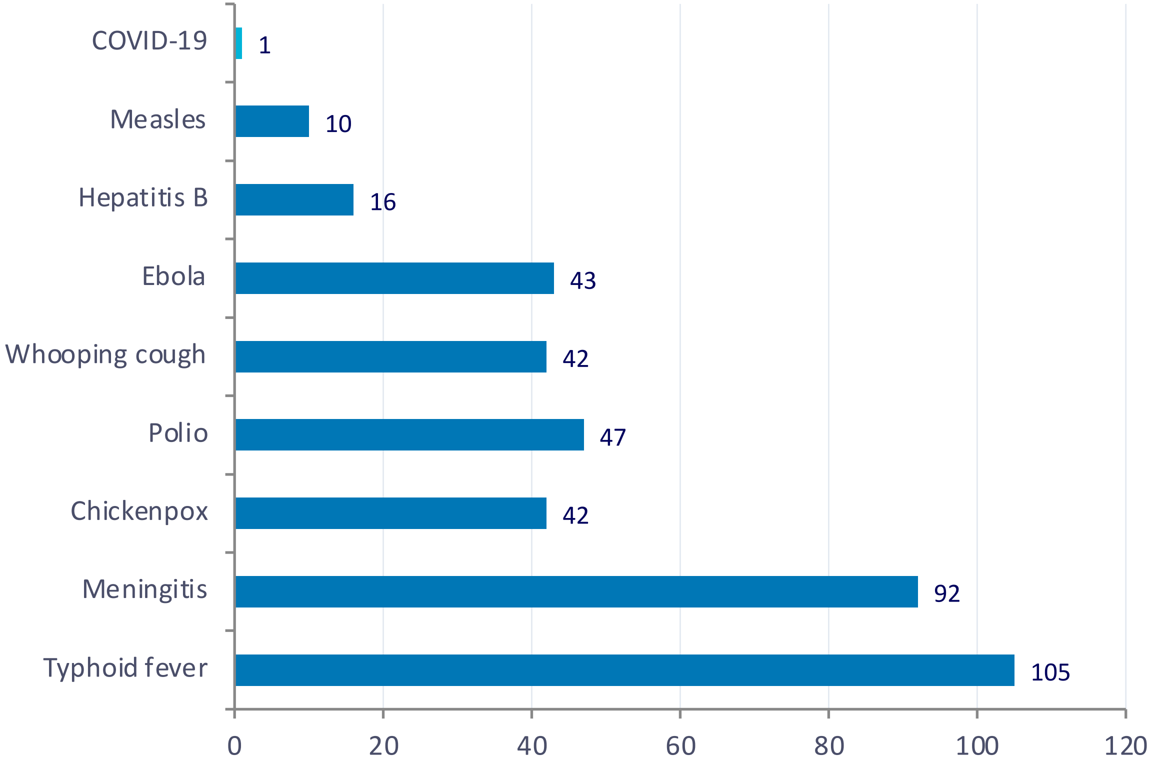
From Smallpox Eradication to COVID-19 in Record Time

Source: Our World in Data | 1796–2020

1796 – 2020

A data-driven overview of vaccine impact

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



<1
year
COVID-19 development time

105 years

Typhoid fever (1884–1989)

Source: Roush & Murphy (2007) via Our World in Data

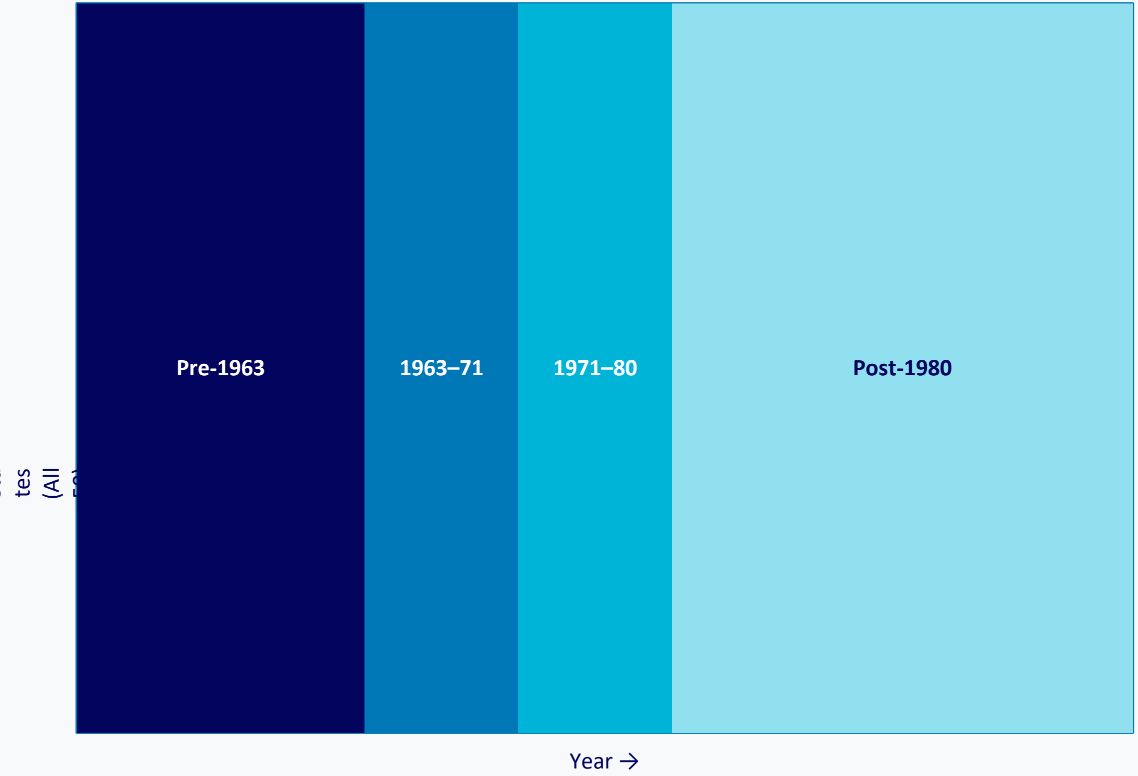
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases	Case Reduction	Death Reduction
Diphtheria	158	0	100% ★	100% ★
Smallpox	250	0	100% ★	100% ★
Acute Polio	141	0	100% ★	100% ★
Measles	3,044	0.2	99.99%	100% ★
Rubella	242	0.04	99.98%	100% ★
Pertussis	1,534	52	96.6%	99.7%
Hepatitis A	465	51	89%	88.7%
Acute Hepatitis B	273	44	83.9%	83.6%
Pneumococcal	233	139	40.5%	31.3%

★ = 100% elimination achieved

Source: Roush & Murphy (2007) via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Incidence level: 100–1,000+/100k | Moderate | Low | Near zero

1963 First measles vaccine developed by John Enders

1971 MMR vaccine introduced by Maurice Hilleman

1980 Vaccination mandatory for kindergarten entry

Before 1963: 100–1,000+ cases per 100,000. After 1980: near zero across all states.

Source: Our World in Data | US Measles 1929–2022

Smallpox: The Only Human Disease Eradicated Through Vaccination

1959
WHO Resolution to Eradicate

1967
Intensified Campaign Launched

1977
Last Natural Case — Somalia

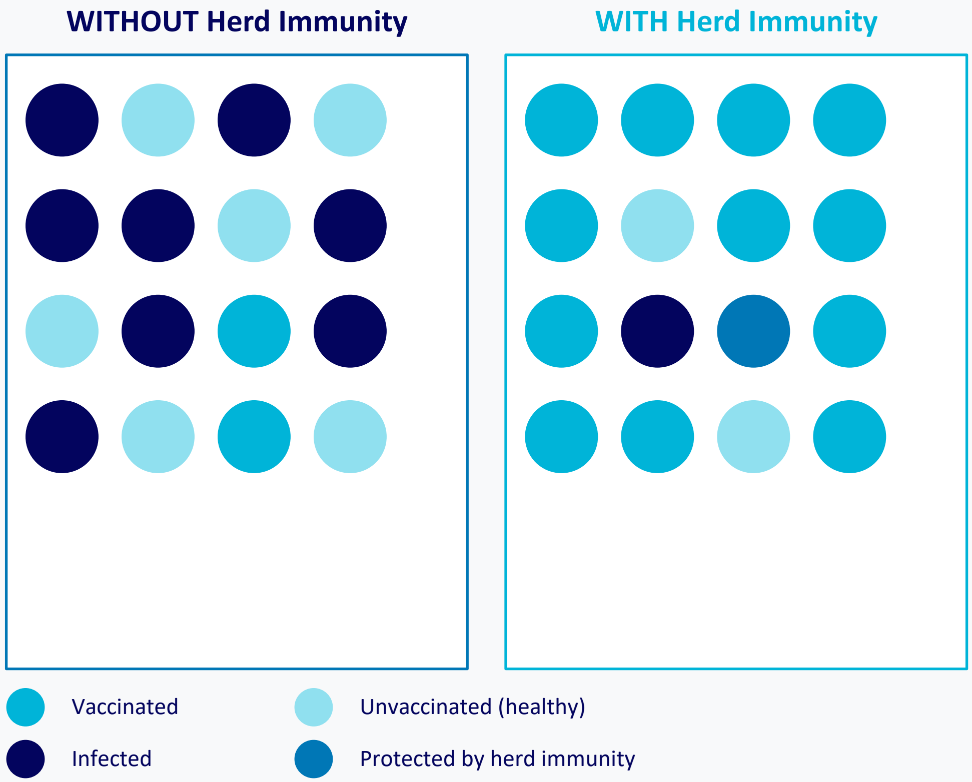
1980
WHO Certifies Eradication

Why Smallpox Was Uniquely Suited for Eradication:

Human-Only Transmission	Visible Symptoms	Cheap & Effective Vaccine
Virus infected only humans	Easily distinguishable from other diseases	Affordable global deployment possible

Source: WHO / Our World in Data

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Source: Our World in Data

Who Is Protected?

Newborns too young to vaccinate, immunocompromised individuals, and cancer patients on chemotherapy cannot receive vaccines — herd immunity shields them.

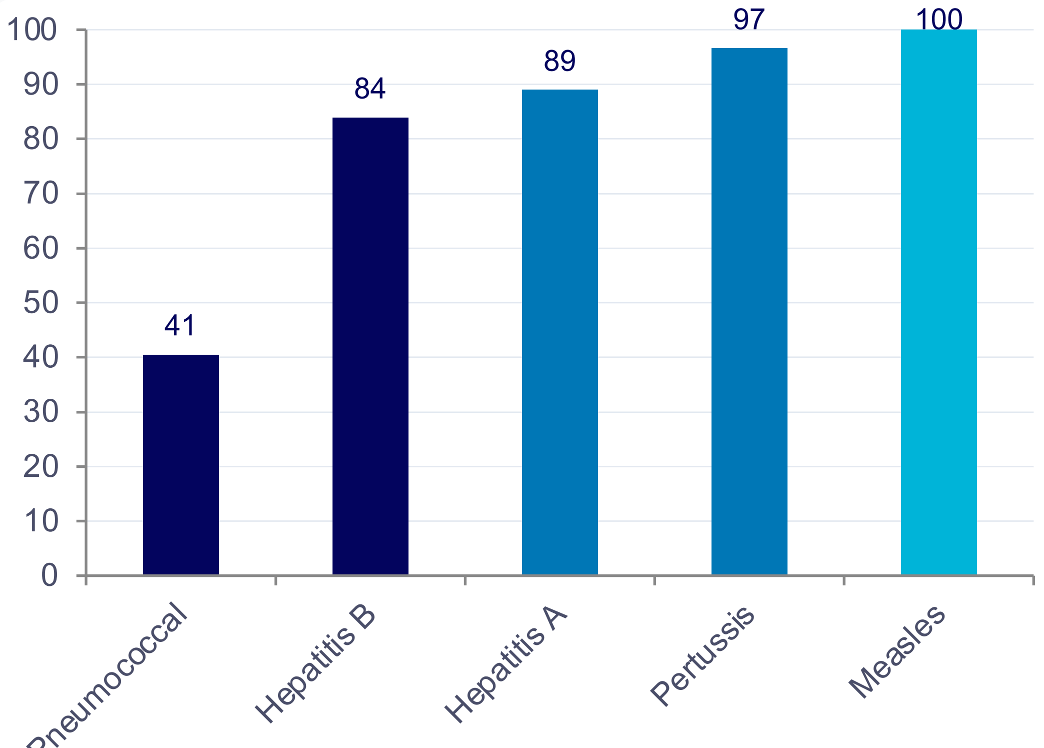
How It Works

When enough people are vaccinated, pathogens cannot spread effectively through the population, breaking transmission chains.

The Result

Entire communities gain protection — especially the most vulnerable — even without direct vaccination.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



100% = complete elimination | Diphtheria, Polio, Smallpox already at 100%

Access Gaps

Not everyone has access to vaccines globally — supply shortages put children at risk.

Supply Shortages

Supply chain constraints leave vulnerable populations at risk, especially in low-income countries.

Trust & Uptake

Poor scientific understanding can reduce trust and vaccine uptake in communities.

Pneumococcal: only 40.5% case reduction — the lowest of all measured diseases

Source: Roush & Murphy (2007) via Our World in Data

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

KEY FINDINGS

- **100% Elimination** achieved for diphtheria, smallpox, and polio in the US
- **Development Timelines Collapsed** from 105 years (typhoid) to under 1 year (COVID-19)
- **Measles Near Zero** across all 50 US states following the 1980 kindergarten vaccination mandate
- **Smallpox Eradicated** — the only human disease eliminated through vaccination, certified 1980
- **Coverage Gaps Remain** — pneumococcal (40.5%), hepatitis A (89%), and hepatitis B (83.9%)

CALL TO ACTION

1. Strengthen global vaccine supply chains and equitable access
2. Invest in next-generation vaccines for pneumococcal, hepatitis, and emerging pathogens
3. Build public trust through evidence-based communication and scientific education

Vaccination remains humanity's most powerful public health tool.

Source: Our World in Data | ourworldindata.org/vaccination